

Phenomenological Fidelity Index (PFI)

Evaluation Rubric for Workflow Comparison Study

Meals & Waier (in preparation)

Purpose

The Phenomenological Fidelity Index (PFI) is a composite measure that provides a structured indicator of how well an analytic output, at a given stage, produced by a given workflow, preserves the phenomenological quality of the original participant data. It is designed for use by blinded, independent reviewers evaluating anonymized stage-level outputs across three workflows: human-only, human-machine, and machine-only.

Justification for a Composite Index

The five PFI dimensions are analytically distinct but not independent:

- 1) **Experiential grounding** establishes the evidentiary floor from which all other interpretive choices must proceed;
- 2) **Phenomenological sensitivity** determines whether that grounding produces genuine meaning-work or merely an accurate summary;
- 3) **Interpretive coherence** governs whether individual meaning-shifts accumulate into a stable account;
- 4) **Structural synthesis** assesses whether the analytic trajectory is genuinely cumulative rather than episodic;
- 5) **Conceptual restraint** disciplines the entire process against the pull of pre-existing theoretical vocabulary.

Because these dimensions operate as an interdependent system (i.e., a deficit in one typically propagates into the others), a composite score captures the overall integrity of the analytic pathway as a phenomenological act. For example, a workflow that scores 3 on coherence but 1 on conceptual restraint has produced a logically tidy account that is not a phenomenological analysis. The PFI composite reflects that failure, where isolated dimension scores might obscure it. The composite is therefore not a convenience simplification of multidimensional data but a theoretically motivated summary of a workflow's preservation of the phenomenological character of the interpretive process as a whole.

The rubric dimensions draw on van Manen's (1997, 2014) criteria for phenomenological quality, particularly his insistence that phenomenological text must remain faithful to the lifeworld and resist premature abstraction, and on Lincoln and Guba's (1985) trustworthiness framework, adapted here for stage-level analytic outputs rather than complete studies. Dimensions 4 and 5 are additionally motivated by Ashwin and colleagues (2023, 2025), who identify over-abstraction and conceptual intrusion as systematic tendencies in LLM-generated qualitative analysis. The composite score is the mean of the reviewers' scores across the five dimensions.

Scoring

Each dimension is scored on a 3-point scale (1 = weak, 2 = moderate, 3 = strong). The PFI for a given output is the mean score across all five dimensions (range: 1.0–3.0). Reviewers also produce a brief analytic memo for each output evaluated (see Reviewer Protocol).

Dimension 1: Experiential Grounding

Is the output anchored in the participant's own account?

Grounded in van Manen's (1997) concept of phenomenological validity, the analytic text must retain fidelity to the lifeworld as the participant described it, not as the analyst reconstructed it from external frameworks. An output is experientially grounded when it stays close to the participant's language, specific situations, and described meanings, rather than importing interpretive vocabulary not used by the participant.

Score	Descriptor	Indicators
3 Strong	Output remains consistently anchored in the participant's described experience.	Interpretive claims can be traced to specific participant language or described situations. No meaning is attributed that is not supported by the transcript.
2 Moderate	Output is generally grounded but contains moments of drift away from the participant's account.	Some interpretive claims go slightly beyond what the transcript supports; participant language is occasionally replaced by analyst vocabulary without evident justification.
1 Weak	Output imposes interpretation on the data rather than drawing it from the data.	The analyst's framing dominates; the participant's voice is paraphrased into abstraction; specific situations or language from the transcript are largely absent.

Dimension 2: Phenomenological Sensitivity

Does the output attend to lived meaning rather than only surface content or abstract themes?

This dimension is grounded in van Manen (2014) and Smith, Flowers, and Larkin's (2009) quality criteria for interpretive phenomenological analysis. Phenomenological sensitivity requires that the analyst attend to *how* the participant experienced the phenomenon, its texture, ambivalence, temporality, and embodiment, not merely what events or attitudes it contained. An insensitive output reduces experience to information.

Score	Descriptor	Indicators
3 Strong	Output attends to the experiential character of the data, including ambivalence, embodied or temporal dimensions, and meaning structures, not only to propositional content.	Analyst notices how the participant experienced, not only what they reported. Output would read as phenomenologically informed to a methodologically trained reader.
2 Moderate	Output captures some experiential texture but tends to summarize or extract content at key points.	Themes are present and not inaccurate, but the lived quality of the experience is sometimes flattened into a categorical description.
1 Weak	The output reads as content coding, not as phenomenological analysis.	Analyst identifies topics or attitudes but not experiential meanings. Phenomenological concepts (meaning, givenness, temporality, relational structure) are absent from the interpretive work.

Dimension 3: Interpretive Coherence

Is the account internally consistent across the output?

This dimension combines Lincoln and Guba's (1985) dependability criterion, the analytic process should be logical, traceable, and stable, and van Manen's (2014) requirement that hermeneutic analysis maintain whole-part consistency throughout. An incoherent output contains interpretive contradictions, introduces unresolved tensions between meaning units and themes, or uses key terms inconsistently.

Score	Descriptor	Indicators
3 Strong	Interpretive claims are internally consistent; concepts are used stably; early and late shifts in the output reinforce rather than contradict each other.	A reader can follow the interpretive logic from beginning to end without encountering unresolved contradictions or unexplained shifts in framing.
2 Moderate	The output is largely coherent but contains local inconsistencies that require revision.	A concept introduced early is later used differently; a theme label does not fully capture the meaning units beneath it; one claim somewhat undercuts another without acknowledgment.
1 Weak	Output is internally contradictory or analytically unstable.	Themes conflict with meaning units; interpretive framing shifts without explanation; key terms are defined implicitly in incompatible ways across the output.

Dimension 4: Structural Synthesis

Does the output integrate earlier analytic insights as it progresses toward abstraction?

This is grounded in van Manen's (1997) whole-part reconciliation requirement and Giorgi's (2009) description of the structure of phenomenological description. Later-stage outputs must carry forward the essential features identified in earlier stages, not simply recapitulate them or abandon them in favor of newly introduced ideas. Weak synthesis occurs when the essence statement does not clearly emerge from the themes and meaning units that precede it.

Score	Descriptor	Indicators
3 Strong	The output demonstrably integrates earlier analytic work. Later moves build on, synthesize, or refine earlier ones.	The path from meaning units to themes to synthesis is traceable. Nothing important from earlier stages is dropped without an evident reason.
2 Moderate	The output shows reasonable integration, but some earlier insights are incompletely carried forward or are arbitrarily excluded.	A meaning unit identified as central does not appear in the themes; a theme mentioned early does not appear in the synthesis; integration is present but partial.
1 Weak	Later-stage analysis does not demonstrably draw on earlier stages.	The essence statement or synthesis could have been written without the preceding analytic work. Earlier meaning units or themes appear discarded rather than incorporated.

Dimension 5: Conceptual Restraint

Does the output avoid imposing constructs that are not clearly grounded in the participant's data?

This dimension builds on Giorgi's (2009) requirement that phenomenological analysis bracket external theoretical frameworks and remain governed by the data, and is directly motivated by Ashwin et al.'s (2023, 2025) identification of conceptual intrusion as a systematic failure mode in LLM-assisted qualitative analysis. LLMs trained on large text corpora tend to map participant accounts onto common theoretical categories present in their training data (e.g., belonging, identity formation, resilience) even when the participant's language does not invite those categories. This dimension evaluates whether that tendency has been resisted.

Score	Descriptor	Indicators
3 Strong	Analytic concepts and labels emerge from the data. External theoretical vocabulary is absent or, if present, is explicitly acknowledged and warranted.	Themes and meaning units are named in language close to the participant's own. If a theoretical term appears, it is because the participant's account warrants it, and the analyst notes that warrant.
2 Moderate	Most conceptual moves are data-grounded, but a few instances of unwarranted theoretical imposition are present.	A theme is labeled using theoretical vocabulary that the participant's account does not clearly support, or a concept is imported from the existing literature without acknowledgment.
1 Weak	External constructs dominate the analytic output.	The analyst consistently maps participant accounts onto pre-existing theoretical categories. Participant language is replaced throughout by theoretical vocabulary. The output reads as if it were applying a framework rather than deriving one.

Reviewer Protocol

Recruiting

Reviewers will be experienced phenomenological researchers who are independent of the study overall.

Blinding

Reviewers receive outputs with the workflow identifier removed (i.e., human, hybrid, machine). The participant identifier is retained to allow stage-level and trajectory-level comparisons within a participant.

Evaluation Levels

Reviewers score at two levels:

- **Stage-level.** Evaluate each output against the participant transcript for that stage.
- **Trajectory-level.** After scoring all stages for one participant/workflow combination, write a brief memo (3–5 sentences) describing how the workflow moved across stages: where it stayed grounded, where it abstracted, and whether integration held.

Reviewer Memos

In addition to scores, each reviewer writes a brief memo per output identifying:

- (a) The strongest and weakest dimensions,
- (b) Any moment where the analytic account diverged notably from the transcript, and
- (c) Whether the output reads as phenomenologically informed or as qualitatively competent but phenomenologically shallow.

Inter-Rater Reliability

With 2–3 reviewers, agreement is assessed using intraclass correlation (ICC, two-way mixed, consistency) for the composite PFI score and Krippendorff's α for individual dimension scores. Reviewer memos are analyzed qualitatively as the interpretive layer of the results.

References

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